ASSIGNMENT 6

(1) Show that retract of a formal space is formal.

(2) Let X, Y be simply connected spaces. Show that $X \times Y$ is formal if and only if X and Y are formal.

(3) Let X_i, $i = 1, 2$ be simply connected spaces. Let $\rho_i : M_{X_i} \to H^*(X_i; \mathbb{Q})$ for $i = 1, 2$, be minimal models. Describe the minimal models for $H^*(X_1 \times X_2; \mathbb{Q})$ $H^*(X_1 \vee X_2; \mathbb{Q})$.

(4) Let X, Y be simply connected spaces. Show that $X \vee Y$ is formal if and only if X and Y are formal.

(5) Prove or disprove connected sum of two manifolds is formal if and only if each one is formal.

(6) For a minimal algebra m, we denote by m_k the subalgebra generated by elements of degree $\leq k$. A map between two DGAs $f : A \to B$ is said to be a k-isomorphism if the induced map in cohomology is isomorphism up to degree k and monomorphism at degree $k+1$. With these definitions in mind, show that Let m, m' be two minimal algebras with $m_k \cong m'_k$. If there is a morphism $m_{k+1} \to m'_{k+1}$ which is $(k+1)$-isomorphism, then $m_{k+1} \cong m'_{k+1}$.

Hint:

Consider the diagram

$$\begin{tikzcd} m_k \cong m'_k \arrow[r] \arrow[d] & m_{k+1} \arrow[d, "f"] \\ m'_{k+1} \arrow[r, "id"] \arrow[ur, dashed, "g"] & m'_{k+1} \end{tikzcd}$$

where the bottom horizontal arrow is the identity. The obstructions to finding a lift $m'_{k+1} \to m_{k+1}$ successively lie in the relative cohomology (see Rational Homotopy Theory and Differential Forms by Griffiths and Morgan[Proposition 11.1]) $H^{i+1}(m_{k+1}, m'_{k+1}, V^i)$ where V^i are the degree i generators of m'_{k+1}. If the relative cohomology $H^{i+1}(m_{k+1}, m'_{k+1})$ vanishes for $i + 1 \leq k + 2$, then all the obstructions vanish. Note, we only have to consider i up to $k + 2$, as m_{k+1} is generated by elements of degrees $\leq k + 1$.

Then consider the long exact sequence in cohomology,

$$\cdots \to H^k(m_{k+1}) \to H^k(m'_{k+1}) \to H^{k+1}(m_{k+1}, m'_{k+1}) \to H^{k+1}(m_{k+1}) \to \cdots .$$

Now, from our assumptions on the map f, it follows that $H^{\leq k+2}(m_{k+1}, m'_{k+1})$ vanishes. Therefore, we have a lift g such that fg is homotopic to the identity on m'_{k+1}.

Similarly, consider the diagram

$$\begin{tikzcd} m_k \cong m'_k \arrow[r] \arrow[d] & m'_{k+1} \arrow[d, "g"] \\ m_{k+1} \arrow[r, "id"] \arrow[ur, dashed, "h"] & m_{k+1}. \end{tikzcd}$$

We get a lift h as before, and conclude that gh is homotopic to the identity. Thus, g is cohomologically surjective in all degrees. Therefore, it is a cohomological isomorphism in all degrees.

Since $fg \simeq id$, f is also a cohomological isomorphism.

Both m_{k+1} and m'_{k+1} are minimal algebras, and quasi-isomorphism implies isomorphism. Hence $m_{k+1} \cong m'_{k+1}$.

This gives us the following:

Let m and m' be minimal algebras. A k-isomorphism between m_k and m'_k induces a quasi-isomorphism.